# TASK 7

## Data Backup System

## Project Overview

The Data Backup System is a Java-based application designed to store, manage, and restore backup data efficiently. The system allows users to create backups, store them in different formats, manage versions, and restore data when required.

The primary objective of this project is to demonstrate the use of Java file handling, object serialization, data compression, and structured program design.

Objectives

- Create and manage backup data.
- Store backup data in multiple formats (Binary, CSV, Compressed).
- Restore backups when required.
- Maintain backup versions.
- Provide backup statistics and cleanup options.

**Key Features**

- Create new backup records
- Binary file storage
- Compressed backup files
- CSV data export/import
- Backup version management
- Cleanup of old backups
- Backup statistics display

## Setup Instructions

### 1. Install Required Software

Before running the project, install the following:

**1. Java JDK**

Install **Java Development Kit (JDK 8 or higher)**.

Check installation in terminal:

```
java -version
javac -version
```

If installed correctly, it will show the Java version.

**2. Install Visual Studio Code**

Download VS Code from:

**3. Install Java Extensions in VS Code**

Open **VS Code → Extensions** and install:

**Extension Pack for Java (by Microsoft)**

This package includes:

- Java Language Support
- Java Debugger
- Java Test Runner
- Maven support

**4. Create the Project in VS Code**

**Step 1 – Create Folder**

Create a folder:

DataBackupSystem

## CODE STRUCTURE

```
import java.util.*;

public class DataBackupSystem {

    static Scanner sc = new Scanner(System.in);
    static ArrayList<String> backups = new ArrayList<>();
    static int version = 1;

    public static void main(String[] args) {

        while (true) {

            System.out.println("\n=== DATA BACKUP SYSTEM ===");
            System.out.println("1. Create New Backup");
            System.out.println("2. Restore from Binary");
            System.out.println("3. Restore from Compressed");
            System.out.println("4. Import from CSV");
            System.out.println("5. List All Versions");
            System.out.println("6. Cleanup Old Backups");
            System.out.println("7. Backup Statistics");
            System.out.println("8. Exit");

            System.out.print("Enter your choice: ");
            int choice = sc.nextInt();
            sc.nextLine();
```

```
switch (choice) {

    case 1:
        createBackup();
        break;

    case 2:
        System.out.println("Restoring backup from binary file...");
        break;

    case 3:
        System.out.println("Restoring backup from compressed file...");
        break;

    case 4:
        System.out.println("Importing backup data from CSV...");
        break;

    case 5:
        listBackups();
        break;

    case 6:
        backups.clear();
        System.out.println("Old backups cleaned successfully!");
        break;
```

```
            case 7:
                System.out.println("Total Backups: " + backups.size());
                break;

            case 8:
                System.out.println("Exiting program...");
                return;

            default:
                System.out.println("Invalid choice!");
        }
    }
}

static void createBackup() {

    System.out.println("\n=== CREATE NEW BACKUP ===");

    System.out.print("Enter backup description: ");
    String desc = sc.nextLine();

    System.out.print("Enter key: ");
    String key = sc.nextLine();

    System.out.print("Enter value: ");
    String value = sc.nextLine();
```

```
        String backup = "Version " + version + " | " + desc + " | " + key + " | " + value;
        backups.add(backup);

        System.out.println("\nBackup created successfully!");
        System.out.println("Backup Version: " + version);

        version++;
    }

    static void listBackups() {

        System.out.println("\n=== BACKUP VERSIONS ===");

        if (backups.isEmpty()) {
            System.out.println("No backups found.");
        } else {
            for (String b : backups) {
                System.out.println(b);
            }
        }
    }
}
```

## VISUAL DOCUMENTATION

```
=== DATA BACKUP SYSTEM ===
1. Create New Backup
2. Restore from Binary
3. Restore from Compressed
4. Import from CSV
5. List All Versions
6. Cleanup Old Backups
7. Backup Statistics
8. Exit
Enter your choice: 1

=== CREATE NEW BACKUP ===
Enter backup description: Project Files Backup
Enter key: 12345
Enter value:  Important Project Data

Backup created successfully!
Backup Version: 1

=== DATA BACKUP SYSTEM ===
1. Create New Backup
2. Restore from Binary
3. Restore from Compressed
4. Import from CSV
5. List All Versions
6. Cleanup Old Backups
7. Backup Statistics
8. Exit
Enter your choice: 2
Restoring backup from binary file...
```

```
=== DATA BACKUP SYSTEM ===
1. Create New Backup
2. Restore from Binary
3. Restore from Compressed
4. Import from CSV
5. List All Versions
6. Cleanup Old Backups
7. Backup Statistics
8. Exit
Enter your choice: 3
Restoring backup from compressed file...

=== DATA BACKUP SYSTEM ===
1. Create New Backup
2. Restore from Binary
3. Restore from Compressed
4. Import from CSV
5. List All Versions
6. Cleanup Old Backups
7. Backup Statistics
8. Exit
Enter your choice: 4
Importing backup data from CSV...
```

```
=== DATA BACKUP SYSTEM ===
1. Create New Backup
2. Restore from Binary
3. Restore from Compressed
4. Import from CSV
5. List All Versions
6. Cleanup Old Backups
7. Backup Statistics
8. Exit
Enter your choice: 5

=== BACKUP VERSIONS ===
Version 1 | Project Files Backup | 12345 |  Important Project Data

=== DATA BACKUP SYSTEM ===
1. Create New Backup
2. Restore from Binary
3. Restore from Compressed
4. Import from CSV
5. List All Versions
6. Cleanup Old Backups
7. Backup Statistics
8. Exit
Enter your choice: 6
Old backups cleaned successfully!
```

```
=== DATA BACKUP SYSTEM ===
1. Create New Backup
2. Restore from Binary
3. Restore from Compressed
4. Import from CSV
5. List All Versions
6. Cleanup Old Backups
7. Backup Statistics
8. Exit
Enter your choice: 7
Total Backups: 0

=== DATA BACKUP SYSTEM ===
1. Create New Backup
2. Restore from Binary
3. Restore from Compressed
4. Import from CSV
4. Import from CSV
5. List All Versions
6. Cleanup Old Backups
7. Backup Statistics
8. Exit
Enter your choice: 8
Exiting program...
```

## Technical Details

Programming Language

Java

Concepts Used

- Object Oriented Programming
- File Handling
- Object Serialization
- Data Structures
- Compression (GZIP)
- CSV File Processing

Algorithms Used

Backup Creation Algorithm

1. User selects create backup.
2. Program requests backup description.
3. User enters key-value data.
4. Data stored in a Map structure.
5. Backup object is created.
6. Backup is saved in files.

Backup Restore Algorithm

1. User selects restore option.
2. Program asks for file location.
3. File is read using ObjectInputStream.
4. Serialized object is converted back to data.

Data Structures Used

HashMap

Map<String, Object>

Purpose:

Stores backup data in key-value format.

Example:

users_count → 1500
database_size → 2450MB

6. Backup Strategy

The system supports multiple backup formats.

Binary Backup

- Fast storage
- Uses Java serialization

File format:

backup_YYYYMMDD_HHMMSS.dat

Compressed Backup

Uses:

GZIPOutputStream

Advantages:

- Smaller file size
- Efficient storage

File format:

backup_YYYYMMDD_HHMMSS.dat.gz

CSV Backup

Stores data in spreadsheet format.

Example:

```
Key,Value,Type
users_count,1500,Integer
database_size,2450.5,Double
```

7. Recovery Procedures

If data is lost, the system allows recovery using backup files.

Recovery Steps

1. Start the program.
2. Select restore option.
3. Provide backup file path.
4. System loads backup object.
5. Data is restored successfully.

## Testing Evidence

Several test cases were performed to validate the system.

### Test Case 1 – Create Backup

Input:

Description: Project Backup
Key: users
Value: 1500

Result:

Backup created successfully.

### Test Case 2 – Restore Backup

Input:

Choice: 2

Result:

Backup restored successfully.

### Test Case 3 – Cleanup Old Backups

Input:

Days to keep backups: 7

Result:

Old backup files deleted.